

Rebuttal for CoRL Submission #2468

We thank the Area Chair and reviewers. We respond below to R1–ZQwd, R2–jAyD, and R3–Fcwd. The new evidence directly addresses the requested progress baselines, optimization ambiguity, mask scope, component ablations, and evaluation protocol. We will revise the paper to narrow the claim to an online, tactile-supervised short-horizon contact prior that reduces pre-contact stall.

[AC, R1, R3] Is VPCE only task progress, and how does it compare with phase/hierarchical conditioning? We thank the reviewers and AC for highlighting this valuable comparison. We added phase-4 and normalized progress-20 predictors on the same held-out trajectories. VPCE obtains AUC/F1 of .963/.759, .885/.705, and .946/.735 on SWIPE/WIPE/INSERT, versus .853/.597, .743/.406, and .808/.539 for progress and .670/.000, .666/.000, and .684/.000 for phase. Thus trajectory time explains part of contact timing, but current wrist RGB and robot motion contain additional contact-specific information.

TABLE I
CONTACT PREDICTION AUC/F1 USING PHASE, PROGRESS, OR VPCE.

Task	Phase-4	Progress-20	VPCE
SWIPE	.670/.000	.853/.597	.963/.759
WIPE	.666/.000	.743/.406	.885/.705
INSERT	.684/.000	.808/.539	.946/.735

We additionally evaluated the conditioned policies under the same real-robot SWIPE protocol. Phase-DP succeeds in 12/20 trials (60%), Progress-DP in 14/20 (70%), and VPCE-DP in 28/30 (93.3%). Thus the gain is not only in contact-label prediction: replacing nominal phase/progress with the online VPCE horizon also improves policy execution. This is a controlled comparison within the same diffusion-policy family, not a reproduction of every hierarchical method; we will add Hierarchical Diffusion Policy and CompILE and avoid claiming superiority over all hierarchical policies. Unlike a discrete phase switch, VPCE continuously conditions one action generator with an online, state-dependent 15-step contact horizon.

[R1, R3] Does the teacher transfer, and how much target tactile data is required? We do not claim universal zero-shot transfer. For downstream deployment, we train target-domain teachers using approximately 60 tactile demonstrations per task. The target teachers obtain held-out AUC/F1 .912/.731, .960/.774, and .940/.725 on WIPE/SWIPE/INSERT.

[R2] Are the learned loss weights well posed? We thank R2 for identifying this optimization ambiguity. Because γ and δ are optimized through the same weighted denoising loss, reducing the weight can directly reduce the objective. A cross-task checkpoint audit shows that this behavior is not uniform: at $\bar{c}=1$, the effective weights are approximately 1.005/2.1/4.4 on SWIPE/WIPE/INSERT in both V and VT

policies. Thus SWIPE nearly reaches the lower bound, whereas WIPE and INSERT do not; the strong task dependence nevertheless shows that the learned scale is not a stable or interpretable mechanism. We replace it with the fixed schedule $w(c) = \text{clip}(1 + c, 1, 2)$. Fixed- $\alpha=1$ robot checks obtain 8/10 on SWIPE and 7/10 on WIPE, while the submitted no-weighting variant already obtains 8/10 on SWIPE. The main contact-anticipation result therefore does not depend on learned weighting; we will present fixed weighting only as a secondary training mechanism.

[R2] Does Eq. (9) mask proprioception, and how is free-space motion smooth? The reviewer is correct that the paper’s notation and original post-encoder routing conflated proprioception, gripper width, and tactile features. Although input-level dropout targeted tactile keys, the post-encoder mask could zero the entire low-dimensional tail. We corrected the routing to always retain robot state and gripper width and mask only tactile/contact features. This controlled variant obtains 13/20 on SWIPE and produces more coherent free-space approach motion. Action-chunk prediction and receding-horizon execution provide additional temporal continuity. We will revise Eq. (9) to distinguish f_{prop} from f_{tac} .

[AC, R3] Which components matter, and why retain gating? We thank the Area Chair and R3 and will move the compact component ablation from the appendix into the main text. The submitted SWIPE results are full 9/10, no horizon 2/10, horizon only 0/10, no tactile masking 0/10, no gating 9/10, and no loss weighting 8/10. These results identify future-contact horizon conditioning and contact-adaptive tactile masking as the core success mechanisms, with loss weighting serving as secondary training emphasis. The identical 9/10 result shows that gating is not required for binary success, so we will not present it as a core mechanism. The appendix’s rollout notes nevertheless show that successful no-gating trials entered contact less stably and required more alignment corrections, suggesting a trajectory-quality effect not captured by binary success. Because this effect was not quantified, we will label it qualitative. We will retain gating only as an auxiliary, task-dependent modulation and provide its feature-weight trace for interpretation, without claiming a success-rate improvement.

[R2] Could operator-held props assist the policy? The reader and foam were held only to keep them at a predefined height and orientation within the robot’s reachable range. After rollout start, the holder did not track the end effector, reposition the object, or assist alignment; all methods used the same placement, success criterion, and 60-s timeout. Crucially, this same protocol did not substantially rescue non-VPCE baselines on SWIPE: DP-V and DP-VT achieved only 1/20 and 2/20, versus 17/20 and 18/20 for VPCE-DP-V and VPCE-DP. Thus holding the reader at a reachable pose alone does not explain the method gap; with the predicted contact

horizon, the policy committed smoothly through the final approach into contact and manipulation, directly addressing the pre-contact stall targeted by this work. We further expanded VPCE-DP to 30 trials per task: SWIPE 28/30 (93.3%), WIPE 22/30 (73.3%), and INSERT 18/30 (60.0%).

This setup places the interaction within the wrist view and the 15-step contact horizon; VPCE is intended to regulate imminent contact rather than plan a long-range approach. For longer-horizon tasks, future work can add a wider or additional camera and a coarse approach stage before wrist-based VPCE regulates the final approach-to-contact transition.

[R2, R3] Why exclude tactile input from VPCE, and how are labels and contact directions handled? VPCE intentionally excludes tactile input because its role is to anticipate contact before touch is informative and to permit tactile-free deployment; tactile is used as privileged training supervision. In the full policy, tactile feedback remains available after contact. For the current planar tasks, contact labels use the temporal change of a base-frame z -force proxy. A threshold of 0.095 is used for the source, SWIPE, and INSERT data; WIPE uses 0.0507 due to its different sensor scale. Each threshold is selected once from the training force-change distribution and then fixed. A $0.6\times-1.4\times$ threshold sweep changes the 15-step positive-label rate smoothly (SWIPE $.296 \rightarrow .217$, WIPE $.582 \rightarrow .379$, INSERT $.390 \rightarrow .290$), rather than creating labels only at one tuned value. Contacts with changing normals can instead use $\|\Delta\mathbf{F}\|_2$, local normal/shear changes, hysteresis, or learned change-point labels without changing the VPCE-policy interface.

[R1, R3] What is the main contribution relative to predictive force guidance and tactile representation learning?

We will sharpen the claim: the contribution is the identification of pre-contact stall, tactile-supervised distillation of a compact future-contact horizon into visual-proprioceptive inputs, and evidence that this signal can initiate contact even without tactile sensing at deployment. The individual gates and loss weights are not claimed as the primary novelty. Unstructured tactile encoders such as Sparsh are complementary for representing post-contact feedback; comparing them does not replace the question addressed here—how to act before tactile observations become informative. We will expand related work and state the missing Sparsh/full hierarchical policy comparisons as evaluation limitations.